Fuel Delivery Optimisation

Student learning guide | ICE laboratory | 31 October 2026

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From your first experiment to checking the equations. This guide matches the in-page learning guide and explains all inputs and calculated outputs.

Start here: your first experiment

Load Warm cruise. Review the inputs, then select Run new simulation. Watch the fuel path, pulse width and fuel total. Pause to inspect a result. Resume continues the recorded run. Run new starts at zero and clears learned trim and old history. Editing an input pauses the experiment and updates a preview without advancing virtual time.

Explore the 3D view

Vehicle context places the fuel path in a vehicle with the project Genesis GV70 baseline dimensions. Injector cutaway enlarges one representative injector and chamber. Drag to turn the camera; pinch or scroll to zoom, or use the zoom buttons. Select Fuel source, Rail, Injector or Cylinder to focus on that component. Reset camera returns to the starting view for that mode. The 2D schematic remains available if your device cannot show 3D.

Understand the motion

The amber path represents fuel delivery. The injector needle and spray appear during a slowed illustration of the commanded opening. Displayed flow, pulse width and duty come from the same calculation as the results. The body, internal arrangement, droplets and spray shape are illustrations. They do not calculate pressure waves, fuel level, atomisation or combustion. The piston stays still because the model has no crank-angle output. Pause and the motion switch stop the illustration; reduced-motion preferences are respected.

Calculation step, in everyday terms

A step is one calculation interval. If Calculation step is 0.1 s, ten calculations cover one virtual second. A 30 s run then records 300 intervals. At 0.02 s it records 1500 intervals. A smaller step updates learning and accumulated fuel more often; it does not change the 3D frame rate. Use the same step when comparing trials.

Understand mixture and fuel corrections

Lambda compares the air-fuel mixture with the nominal gasoline balance of 14.7:1 by mass. Lambda 1 is the nominal balance; below 1 is rich and above 1 is lean. STFT is the quick correction and LTFT is the slow learned correction. With target 1 and measured 1.05, STFT is +6%; measured 0.95 gives -6%. LTFT learning gain 0 keeps the current learned value while STFT can still react. Switching closed loop off sets STFT to zero and stops learning without erasing LTFT. Reset or Run new clears it.

Read the graphs and gauges

Graphs show recorded virtual time, not wall-clock time. Each chart names its signal and unit. Solid, dashed and dotted lines distinguish traces as well as colour. Injector duty shows the requested share of the cycle, limited to 100% for display. Fuel consumed is the accumulated mass; its progress bar uses a display scale rather than a tank capacity. A preview after an edit can differ from the last recorded chart point.

Try and verify: the default run

Reset to defaults, keep the 30 s duration and 0.1 s calculation step, then run to completion. Initial pulse width is about 8.471 ms and fuel flow is about 1.701 g/s. After 30 s, fuel totals about 51.020 g and BSFC is about 292.325 g/kWh. Small rounding differences in visible numbers are normal; exports retain more detail. These are software reference checks, not measurements from a real engine.

Try and verify: one input at a time

Reset before each trial. Change air flow from 25 to 50 g/s: preview fuel flow doubles to about 3.401 g/s. Change measured lambda to 1.05 with target 1: the quick correction becomes +6%. Raise rail pressure with all other inputs fixed: pulse width gets shorter. Lower battery voltage: opening delay grows. Set air flow to 200 g/s at the other defaults: duty reaches its display limit and warns that delivery is theoretical.

Compare fuel trials fairly

Compare baseline and adjusted calculates two fresh full-duration runs from the current inputs, with scale as their only difference. It does not replay edits in a recorded run. Apply recommended scale changes the multiplier only on request. Keep duration, step, load, torque and all other inputs the same. Check predicted lambda, target error, duty and warnings alongside fuel totals. Less fuel alone does not establish a better result.

Advanced: how the calculation is connected

Events per second = RPM / 120 x 4. Air per event = air flow x 1000 / event rate. Target air-fuel ratio = 14.7 x target lambda. Base fuel = air per event / target ratio. Warm-up, load and manual acceleration factors form a corrected baseline. STFT and LTFT multipliers, fuel scale and injector bias adjust it. Pressure-corrected injector flow uses the square root of pressure / 300 kPa and density 0.745 g/cc. Pulse width = final event fuel / corrected flow + opening delay.

Advanced: limits and export evidence

Requested duty = pulse width x RPM / 1200 and is displayed between 0 and 100%. Theoretical fuel is not reduced when saturation occurs. Predicted lambda divides the measured input by trims, scale and bias; the measured input is not an automatically generated feedback signal. Fuel is accumulated over the selected time step. The LTFT integrator keeps its internal accumulation even when its displayed output reaches +/-20%. JSON includes named outputs and an eight-value integration vector; CSV includes inputs and outputs for each recorded sample. Save the settings with each result.

Reference evidence and model boundaries

The existing implementation was derived from the supplied programmer guide, implementation script and inspected Fuel_Delivery_Optimisation_All_In_One.slx equations and wiring. No MATLAB numerical run was performed. The GV70 baseline supplies overall dimensions, wheelbase, track and tyre radius; the body is not manufacturer CAD and the fuel-system arrangement is illustrative. This module has four cylinders and a nominal 14.7:1 gasoline ratio. It is an educational software model without hardware connection or OEM calibration. Numerical tests check implementation behaviour; they do not validate fuel economy or real injector spray.

Chemistry context

The nominal gasoline ratio used here is 14.7:1 by mass. An illustrative iso-octane reaction is 2 C8H18 + 25 O2 -> 16 CO2 + 18 H2O. Hydrogen in the fuel combines with oxygen to form water. This chemical example does not replace the model's nominal gasoline ratio or add a combustion calculation.

Every input, in plain language

Suggested ranges guide classroom experiments. Only fuel scale, duration and calculation step have the named hard ranges shown below; other validation rules still apply.

Engine speed (rpm)

How fast the engine turns. More RPM means more injection events each second. It also changes injector duty and calculated engine power.

Default: 2500. Suggested range: 600 to 6000 rpm. Signal: engineSpeedRpm.

Engine load (%)

How hard the engine is working. Above 70% load, this model adds extra fuel. Air flow is still a separate input that you set yourself.

Default: 35. Suggested range: 0 to 100 %. Signal: engineLoadPct.

Air flow (g/s)

How many grams of air enter each second. More air usually needs more fuel to keep the same mixture. This value does not change automatically with load.

Default: 25. Suggested range: 2 to 200 g/s. Signal: airFlowGps.

Coolant temperature (°C)

How warm the engine coolant is. Below 80 °C, the model adds warm-up fuel. At 20 °C or colder it adds the maximum 25% extra.

Default: 90. Suggested range: -10 to 110 °C. Signal: coolantTempC.

Brake torque (N m)

The turning force at the engine shaft. It helps calculate power and fuel used per unit of work (BSFC). It does not directly change the fuel command.

Default: 80. Suggested range: 0 to 300 N m. Signal: brakeTorqueNm.

Target lambda (ratio)

The mixture you want. A value of 1 means the nominal balance of air and fuel. Below 1 means richer (more fuel); above 1 means leaner (less fuel).

Default: 1. Suggested range: 0.85 to 1.1 ratio. Signal: targetLambda.

Measured lambda (ratio)

The mixture reading you choose for the virtual sensor. It is independent of the calculated mixture. Above target asks for more fuel; below target asks for less.

Default: 1. Suggested range: 0.8 to 1.2 ratio. Signal: measuredLambda.

Closed-loop fuel control

On: the controller adjusts fuel using the sensor reading and slowly learns a correction. Off: the immediate correction stops, but the learned correction is kept.

Default: True. Suggested range: 0 to 1 . Signal: closedLoopEnabled.

LTFT learning gain (1/s)

How quickly the slow fuel correction learns. Zero freezes the learned value; the quick correction can still change. A larger value learns faster.

Default: 0.003. Suggested range: 0 to 0.01 1/s. Signal: ltftLearningGainPerS.

Fuel-rail pressure (kPa)

The pressure supplying the injector. More pressure lets the injector pass fuel faster, so the same fuel amount needs a shorter opening time.

Default: 300. Suggested range: 200 to 400 kPa. Signal: fuelRailPressureKPa.

Battery voltage (V)

The electrical supply to the injector. Lower voltage makes the injector take longer to open. That extra delay is included in the pulse width.

Default: 14. Suggested range: 10 to 15 V. Signal: batteryVoltageV.

Injector rated flow (cc/min)

How much fuel the injector is rated to pass in one minute at 300 kPa. A larger injector can deliver the same fuel amount in less time.

Default: 220. Suggested range: 150 to 500 cc/min. Signal: injectorRatedFlowCcMin.

Injector flow bias (%)

A delivery error you choose. Positive values mean more fuel is delivered than commanded. Negative values mean less fuel is delivered.

Default: 0. Suggested range: -20 to 20 %. Signal: injectorFlowBiasPct.

Acceleration enrichment (%)

Extra fuel you add to represent a sudden increase in demand. You set it manually; the model does not calculate vehicle acceleration.

Default: 0. Suggested range: 0 to 30 %. Signal: accelerationEnrichmentPct.

Fuel optimisation scale (ratio)

A multiplier for testing the fuel amount. 1 keeps it unchanged; 0.95 requests 5% less. Check the mixture and warnings before calling a trial better.

Default: 1. Allowed range: 0.85 to 1.05 ratio. Signal: fuelOptimisationScale.

Simulation duration (s)

How many virtual seconds the run will last. For example, 30 means half a minute in the experiment. Reset before reducing it below the time already recorded.

Default: 30. Allowed range: 5 to 120 s. Signal: simulationDurationS.

Calculation step (s)

The gap between calculations in virtual seconds. At 0.1 s, the model calculates ten times per virtual second. It is separate from animation speed. Smaller steps give more frequent samples.

Default: 0.1. Allowed range: 0.02 to 1 s. Signal: timeStepS.

Every calculated output

Injection events per second

How many injection events occur each second across the four-cylinder, four-stroke model.

Air mass mg per event

Air amount assigned to one injection event.

Target afr

Desired air mass divided by fuel mass. At lambda 1 the model uses 14.7 parts air to 1 part fuel.

Base fuel mg per event

Fuel needed for one event before warm-up, load and other adjustments.

Warmup factor

Multiplier that adds fuel while coolant is cold.

Load factor

Multiplier that adds fuel at high load.

Acceleration factor

Multiplier for the extra fuel you manually request.

Baseline corrected fuel mg per event

Fuel after warm-up, load and acceleration adjustments, before trims, scale and injector bias.

Lambda error

Measured lambda minus target lambda.

Stft pct

Short-term fuel trim: the quick correction. Positive adds fuel; negative removes it.

Ltft pct

Long-term fuel trim: the slow correction learned over time.

Fuel trim factor

Combined effect of the quick and slow fuel corrections.

Recommended fuel scale

A suggested multiplier from measured lambda divided by target lambda, limited to 0.85-1.05. It is applied only when you request it.

Commanded fuel mg per event

Requested fuel per event after trims and the selected scale.

Final fuel mg per event

Theoretical fuel per event after injector bias is included.

Corrected injector flow mg per ms

Fuel the injector can pass per millisecond after pressure correction.

Injector dead time ms

Electrical opening delay included in the pulse width.

Injector pulse width ms

Total commanded injector time in milliseconds, including the opening delay.

Raw duty cycle pct

Requested injector duty before the display limit of 100% is applied.

Injector duty cycle pct

Percentage of the available cycle requested for injection. Above 85% raises a warning; at 100% the injector is shown as saturated.

Predicted lambda

Estimated mixture after trims, scale and injector bias. This is a calculated estimate, not a new sensor measurement.

Fuel saving pct

Percentage reduction from the untrimmed corrected-fuel baseline. It can be negative and does not prove a safe or efficient optimum.

Fuel flow gps

Theoretical fuel used each second, in grams.

Fuel flow lph

Theoretical fuel flow expressed in litres per hour.

Cumulative fuel g

Total fuel counted during the recorded run, in grams.

Brake power kw

Engine output power calculated from torque and RPM.

Bsfc gper kwh

Fuel needed for one unit of engine work. Lower can be better in a comparable valid test. This value is not meaningful near zero power.